

Test Results

Patient: Michael Johnson | DOB: 1978-11-03 | Patient ID: P002

Collected: 2022-06-20 | Source: Patient portal sample export

COMPREHENSIVE METABOLIC PANEL (CMP)

Results

Sodium

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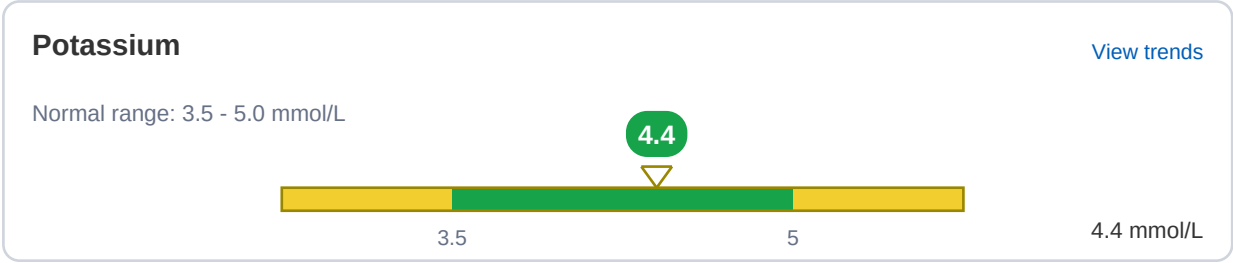
Normal range: 135 - 145 mmol/L



Potassium

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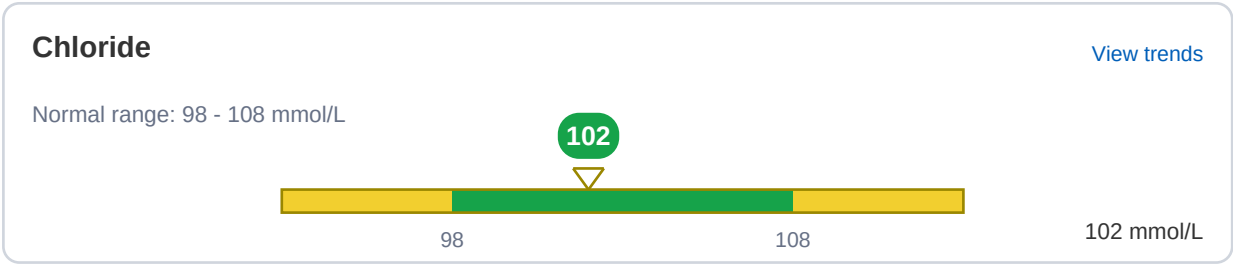
Normal range: 3.5 - 5.0 mmol/L



Chloride

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Normal range: 98 - 108 mmol/L



Carbon Dioxide (CO2)

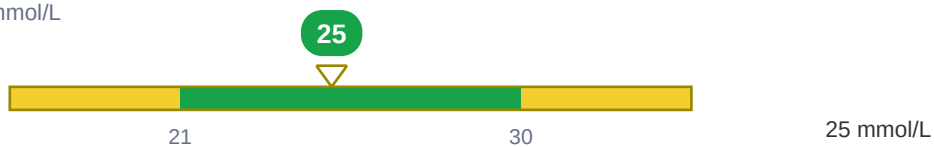
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Normal range: 21 - 30 mmol/L

Carbon Dioxide (CO₂)

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Normal range: 21 - 30 mmol/L



Urea Nitrogen (BUN)

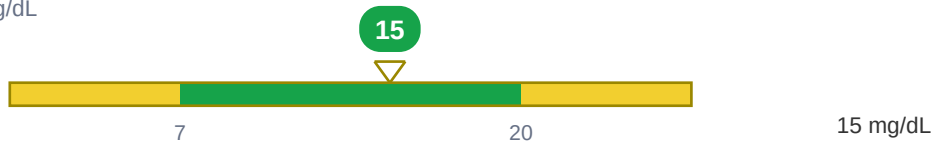
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Normal range: 7 - 20 mg/dL

Urea Nitrogen (BUN)

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Normal range: 7 - 20 mg/dL



Creatinine

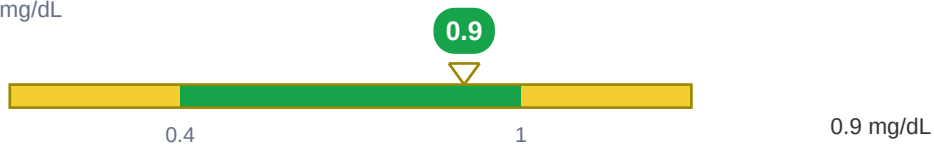
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Normal range: 0.4 - 1.0 mg/dL

Creatinine

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Normal range: 0.4 - 1.0 mg/dL



Glucose

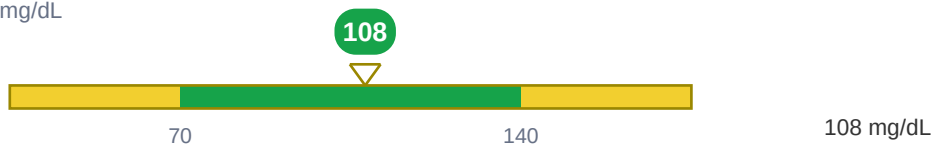
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Normal range: 70 - 140 mg/dL

Glucose

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Normal range: 70 - 140 mg/dL



Interpretive Data:

Above is the NONFASTING reference range.

Below are the FASTING reference ranges:

NORMAL: 70-99 mg/dL

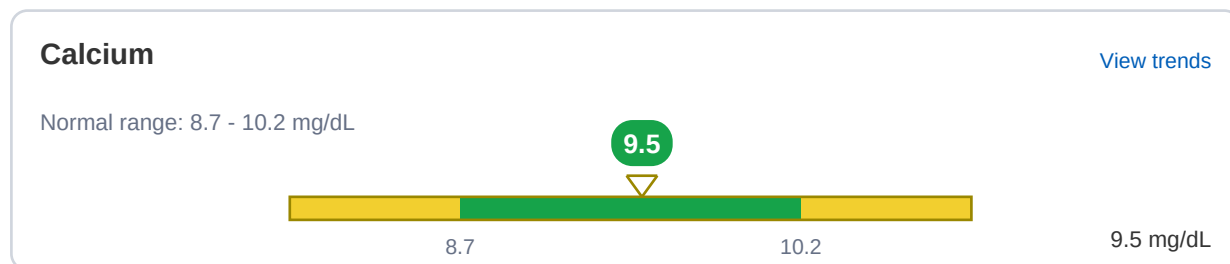
PREDIABETES: 100-125 mg/dL

DIABETES: > 125 mg/dL

Calcium

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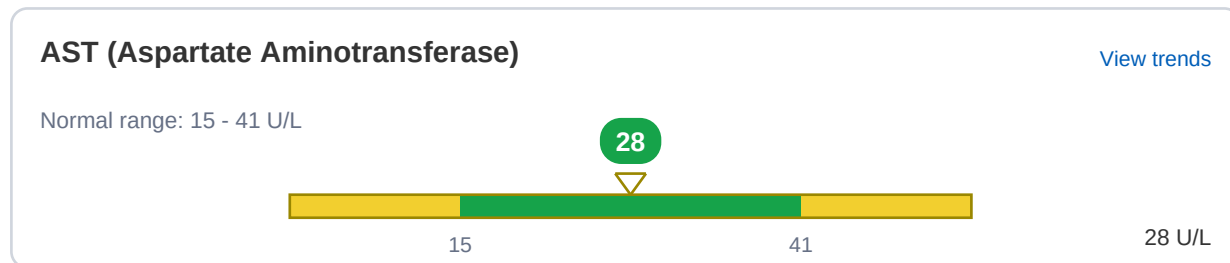
Normal range: 8.7 - 10.2 mg/dL



AST (Aspartate Aminotransferase)

[View trends](#)

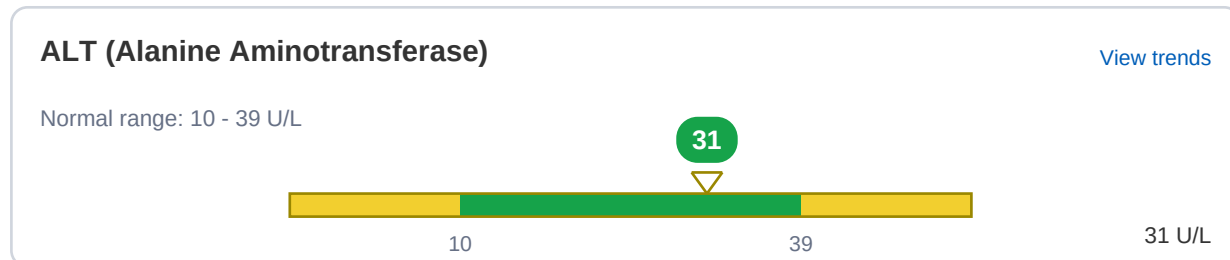
Normal range: 15 - 41 U/L



ALT (Alanine Aminotransferase)

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Normal range: 10 - 39 U/L



Bilirubin, Total

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Normal range: 0.4 - 1.5 mg/dL

Bilirubin, Total

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Normal range: 0.4 - 1.5 mg/dL



0.9 mg/dL

Alk Phos (Alkaline Phosphatase)

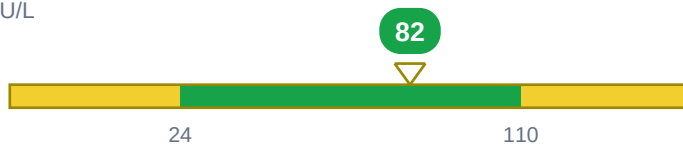
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Normal range: 24 - 110 U/L

Alk Phos (Alkaline Phosphatase)

[View trends](#)

Normal range: 24 - 110 U/L



82 U/L

Albumin

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Normal range: 3.5 - 4.8 g/dL

Albumin

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Normal range: 3.5 - 4.8 g/dL



4.2 g/dL

Protein, Total

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Normal range: 6.2 - 8.1 g/dL

Protein, Total

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Normal range: 6.2 - 8.1 g/dL



7.2 g/dL

Anion Gap

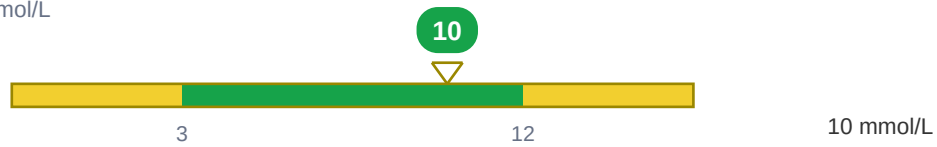
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Normal range: 3 - 12 mmol/L

Anion Gap

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Normal range: 3 - 12 mmol/L



BUN/CREA Ratio

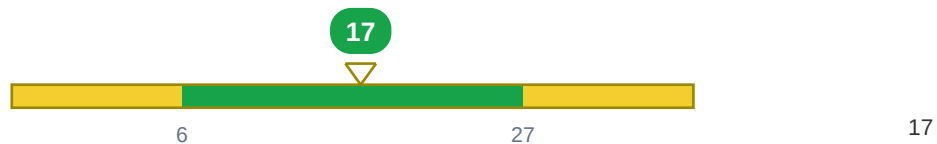
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Normal range: 6 - 27

BUN/CREA Ratio

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Normal range: 6 - 27



Glomerular Filtration Rate (eGFR)

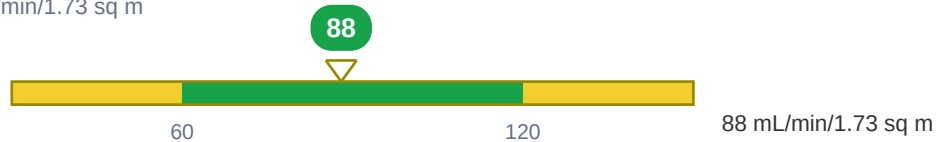
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Normal range: > 60 mL/min/1.73 sq m

Glomerular Filtration Rate (eGFR)

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Normal range: > 60 mL/min/1.73 sq m



CKD-EPI (2021) does not include patient's race in the calculation of eGFR. Monitoring changes of plasma creatinine and eGFR over time is useful for monitoring kidney function. This change was made on 3/1/2022.

Interpretive Ranges for eGFR(CKD-EPI 2021):

eGFR: > 60 mL/min/1.73 sq m - Normal
eGFR: 30 - 59 mL/min/1.73 sq m - Moderately Decreased
eGFR: 15 - 29 mL/min/1.73 sq m - Severely Decreased
eGFR: < 15 mL/min/1.73 sq m - Kidney Failure

Note: These eGFR calculations do not apply in acute situations when eGFR is changing rapidly or in patients on dialysis.

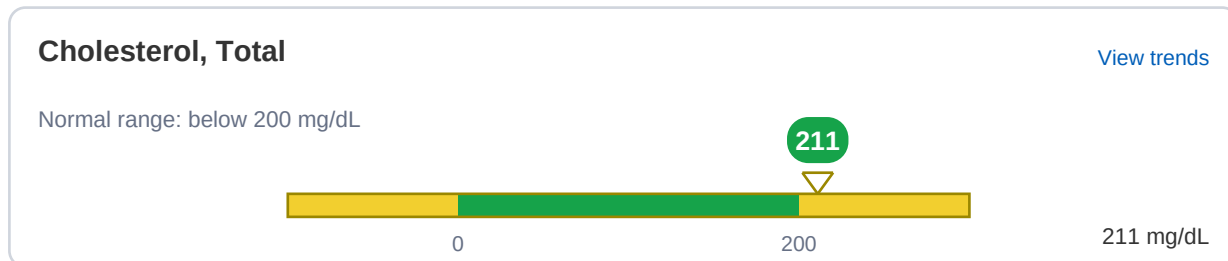
Lipid panel:

Results

Cholesterol, Total

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Normal range: below 200 mg/dL



Comment: The significance of total cholesterol depends on the values of individual components including HDL, LDL, non-HDL, and triglycerides. A fasting or non-fasting lipid panel, including total, HDL, and LDL cholesterol, and triglycerides, is recommended to screen and monitor lipid-related risk.

Desirable: <200 mg/dL

Borderline High: 200-239 mg/dL

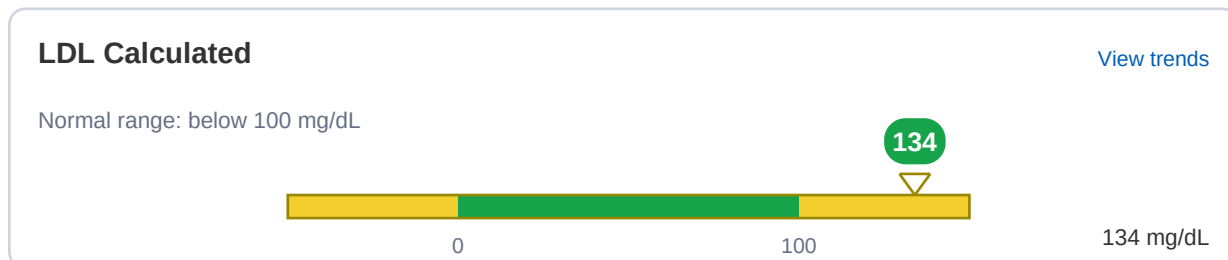
High: ≥240 mg/dL

Ref: NCEP guidelines

LDL Calculated

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Normal range: below 100 mg/dL



<70 mg/dL Desired target for prior heart disease, stroke, and those at high-risk. Even lower levels may be recommended to decrease the risk of heart attack and stroke.

70-159 mg/dL Comprehensive cardiovascular risk assessment is recommended. Statin therapy may be advised based on risk factors.

Desirable: <100 mg/dL

Above desirable: 100-129 mg/dL

Borderline high: 130-159 mg/dL

160-189 mg/dL Moderately elevated LDL level. Statin therapy is recommended if other risk factors are present.

≥190 mg/dL Severely elevated LDL level. High long-term risk of heart disease and stroke. High-intensity statin therapy is recommended for most people. Consider specialist referral.

*A healthy diet and exercise are recommended for all to reduce heart disease risk. Statin choice should be based on patient preference after patient-provider discussions.

Ref: 2018 ACC/AHA and the NCEP guidelines

HDL

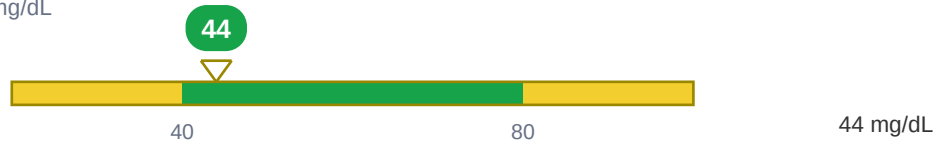
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Normal range: 40 - 80 mg/dL

HDL

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Normal range: 40 - 80 mg/dL



People with low HDL levels are at increased risk of heart disease:

<50 mg/dL for Women

<40 mg/dL for Men

Triglyceride

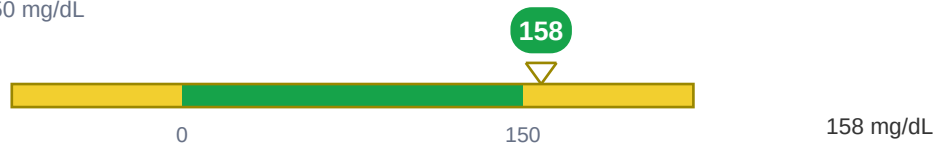
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Normal range: below 150 mg/dL

Triglyceride

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Normal range: below 150 mg/dL



<150 mg/dL Normal

150-499 mg/dL High Triglycerides. Risk of heart disease may be increased. Address reversible causes (eg sugar in foods and beverages, alcohol, and diabetes control). Medication may be appropriate based on other clinical factors.

>=500 mg/dL Very High Triglycerides. Risk of heart disease and pancreatitis increased. Address reversible causes as above. Medication to lower triglycerides usually advised.

*Ranges provided for adults, pediatric guidelines vary.

VITAMIN D 25 HYDROXY

Results

Bone and Mineral Metabolism: Vitamin D

25 OH Vitamin D Status

<10 ng/mL Deficiency

10-30 ng/mL Insufficiency

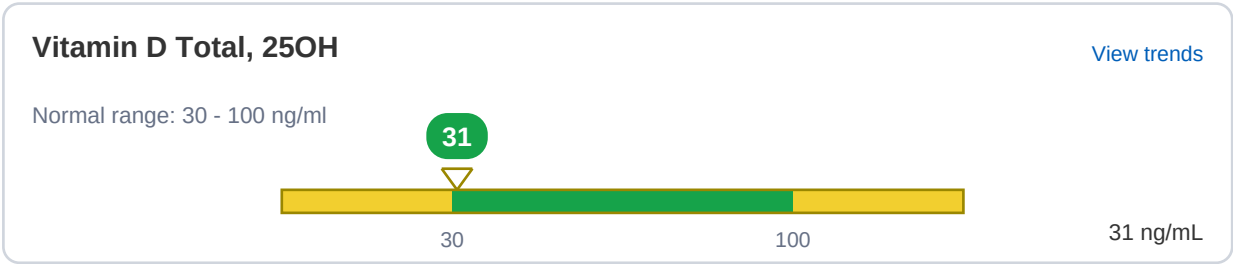
30-100 ng/mL Sufficiency

>100 ng/mL Toxicity

Vitamin D Total, 25OH

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Normal range: 30 - 100 ng/ml



HEMOGLOBIN A1C

Results

Between 5.7% and 6.4% is suggestive of Pre-Diabetes or controlled Diabetes.

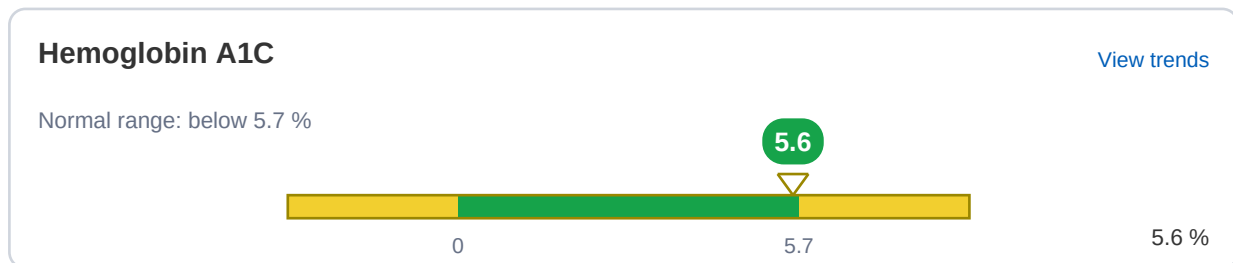
Greater than or equal to 6.5% is suggestive of Diabetes, and if more than one value, diagnostic.

Accuracy may be reduced by anemia, hemoglobinopathy, recent transfusion, sickle cell, artificial heart valve, dialysis, TIPS, severe hyperglycemia, etc.

Hemoglobin A1C

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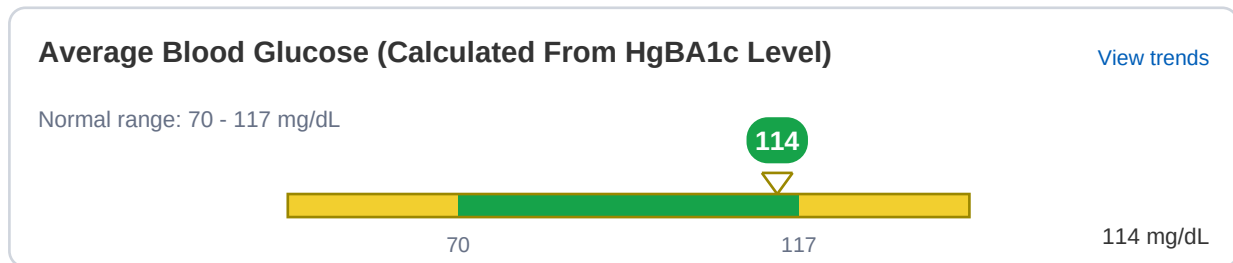
Normal range: below 5.7 %



Average Blood Glucose (Calculated From HgBA1c Level)

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Normal range: 70 - 117 mg/dL



Thyroid Hormone Tests

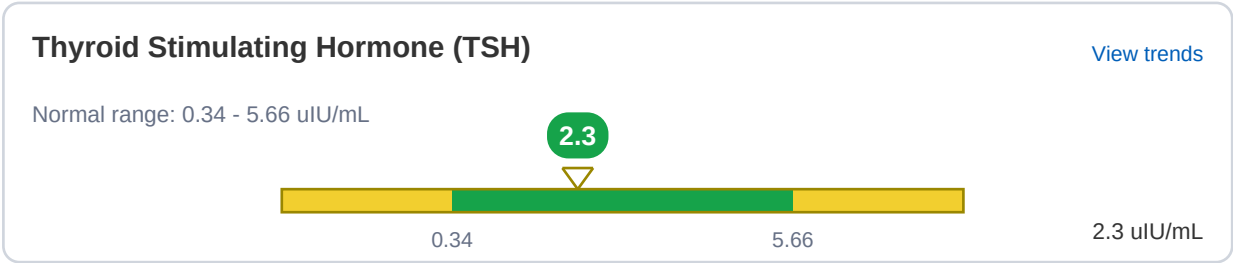
Collected on Jun 17, 2024 9:38 AM

Results

Thyroid Stimulating Hormone (TSH)

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Normal range: 0.34 - 5.66 uIU/mL



Thyroxine, Free (FT4)

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Normal range: 0.52 - 1.21 ng/dL



VITAMIN B12

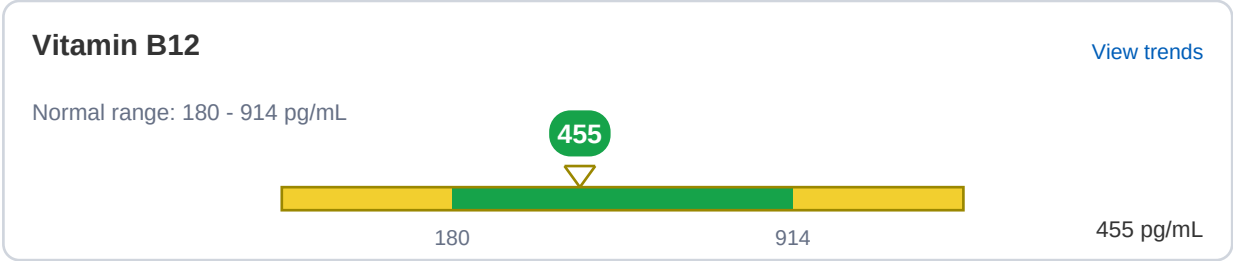
Collected on Jun 17, 2024 9:38 AM

Results

Vitamin B12

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Normal range: 180 - 914 pg/mL



IRON AND TOTAL IRON BINDING CAPACITY (TIBC)

Collected on Jun 17, 2024 9:38 AM

Results

Iron

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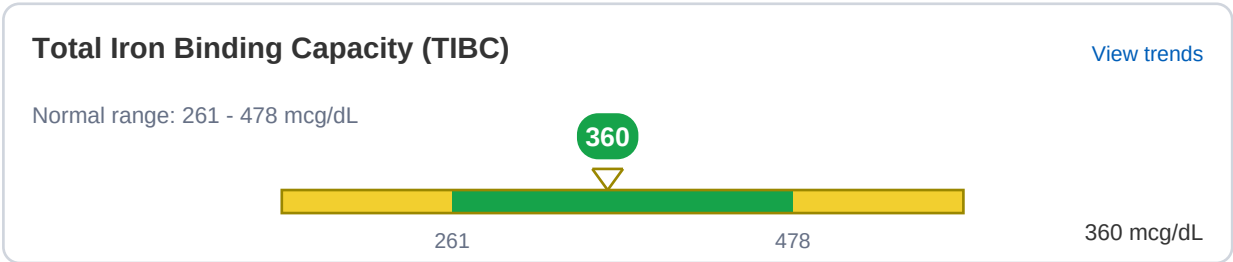
Normal range: 28 - 170 mcg/dL



Total Iron Binding Capacity (TIBC)

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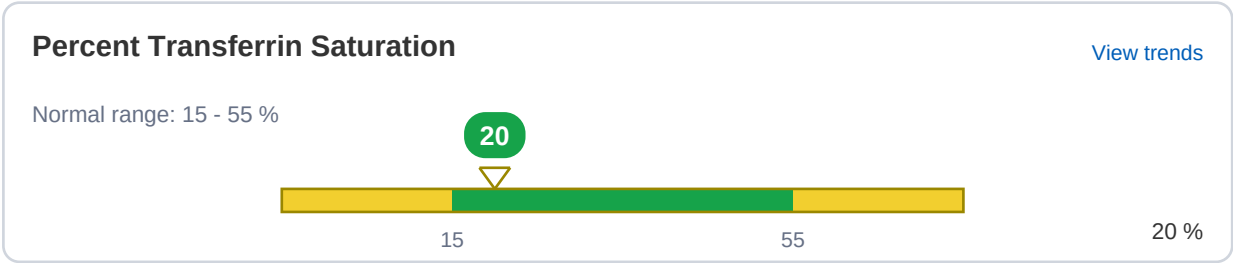
Normal range: 261 - 478 mcg/dL



Percent Transferrin Saturation

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Normal range: 15 - 55 %



Additional information

Synthetic sample data for software testing only. Not a real medical record.